Bench exercise: lines that agree

Japanese Joinery by Hand · Episode 2 · Part 2. Snap a line, square it around a board, and gauge a parallel, then check that all of them close up. You can use a *sumitsubo* 墨壺, *sashigane* 差金 and *kebiki* 罫引, or their Western cousins: a chalk line, a square and a marking gauge. About 40 minutes.

> **Safety.** A marking knife and a gauge blade are sharp. Cut away from your hand, keep your fingers behind the edge, and put the knife down blade-in when you're done.

1 · Snap a straight line

1. On a board at least 600 mm long, push the pin in at one end, close to the edge.
2. Draw the line out to the other end and hold it down, taut.
3. Lift it straight up, a few centimeters, square to the face, and let go.
4. **Check:** lay a straightedge along it. Then snap a second line, lifting the string *sideways* on purpose. Compare the two: which one is straight?

2 · Square a line around the board

1. Pick one edge as the reference and mark it. It must be straight and true; plane it or check it first.
2. Hook one arm of the square against that edge. Press near the middle of the hooked arm and draw along the other arm across the face.
3. Carry the line around: down the edge, across the back, and up the far edge. Always hook the square against a **reference** face or edge.
4. **Check:** the last line should meet the first exactly at the corner. If it misses, check the square, the reference edge, and your grip, then try again.

3 · Gauge a parallel line

1. Set the gauge 30 mm from the fence to the blade, and lock it.
2. Run the fence along the reference edge. Keep it pressed to the edge and score lightly; go over the line twice rather than pressing hard once.
3. Gauge a second board with the **same** setting.
4. **Check:** stack the boards with their reference edges flush. The two lines should sit exactly on top of each other. That is the point of a gauge: set it once, and repeat it on every piece.

4 · Knife line vs pencil line

1. Square one pencil line and one knife line across the grain.
2. Look at both under a lamp, and at an angle. Note how wide each one is, and where exactly its edge is.
3. **Check:** drop a chisel corner or a pencil point into the knife line. It finds the line; on the pencil line, it doesn't.

What you should have

- One straight snapped line, plus one bowed line you made on purpose.
- A line that goes around four faces and closes.
- Two boards with matching gauged lines.

Japanese tools are not required; the ideas are the same. Grip and technique for sharp tools are best learned from a teacher, in person.